



HK IMO Training 2021-2022
Problem Set 16
For 29 January 2022



Problem 61. We colour every point on the plane red or blue.

- (a) Is there a colouring such that there is exactly one red point on each unit circle?
- (b) Is there a colouring such that there are exactly two red points on each unit circle?

Problem 62. Let $\triangle ABC$ be an acute triangle with circumcentre O . Let D be the point on the perpendicular bisector of AC such that $DC \perp BC$. Let E be the point on the perpendicular bisector of AB such that $EB \perp BC$. Let P be an arbitrary point on the side BC . The line passing through P and perpendicular to BO meets the side AB at X . The line passing through P and perpendicular to CO meets the side AC at Y . Prove that the circumcentre of $\triangle AXY$ lies on the line DE .

Problem 63. Let x, y, z be positive real numbers satisfying $\frac{x}{y} + \frac{y}{z} + \frac{z}{x} = 5$. Suppose $\sqrt{x}, \sqrt{y}, \sqrt{z}$ are the side lengths of a triangle. Prove that

$$\frac{x(y^2 - 2z^2)}{z} + \frac{y(z^2 - 2x^2)}{x} + \frac{z(x^2 - 2y^2)}{y} \geq 0.$$

Problem 64. Let k be a positive integer. A positive integer n is *nice* if n is the sum of square of exactly k integers divisible by k . A positive integer m is *funny* if m is the sum of square of exactly k integers not divisible by k . Find all positive integers k such that every nice integer is funny.